Serii de funcții

Studiați convergența simplă și uniformă.

1. $f_n : (0,1) \to \mathbb{R} : f_n(x) = \frac{1}{nx+1} ; n \geq 0$

c. s. $\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} \frac{1}{nx+1} = 0 = f(x)$.

c. u. $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in(0,1)} |f_n(x) - f(x)| \right)$.

Notăm: $g(x) = |f_n(x) - \underbrace{f(x)}_{0}| = \frac{1}{nx+1} ; g : (0,1) \to \mathbb{R}$

$g'(x) = \frac{-n}{(nx+1)^2} < 0 ; \forall x \in (0,1) ; n \geq 0$

x	0	$\qquad$	1
$g'(x)$	$-$	$-\ -\ -\ -\ -\ -\ -\ -\ -\ -$	$-$
$g(x)$	1	$\searrow$	$\frac{1}{n+1}$

$\Rightarrow \sup\limits_{x\in(0,1)} g(x) = \lim\limits_{x\to 0} g(x) = \lim\limits_{x\to 0} \frac{1}{nx+1} = 1$

$\Rightarrow \lim\limits_{n\to\infty} \left(\sup\limits_{x\in(0,1)} g(x) \right) = 1 \neq 0 \Rightarrow$ nu e uniform convergență.

2. $f_n : [0,1] \to \mathbb{R} : f_n(x) = nx(1-x)^n ; n \geq 0.$

c. s. $\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} nx(1-x)^n$

$\lim\limits_{n\to\infty} \frac{f_{n+1}(x)}{f_n(x)} = \lim\limits_{n\to\infty} \frac{(n+1)x(1-x)^{n+1}}{nx(1-x)^n} = \lim\limits_{n\to\infty} \frac{n+1}{n} \cdot (1-x) =$

$= 1-x < 1 ; \forall x \in (0,1]$

$\Rightarrow \lim\limits_{n\to\infty} f_n(x) = 0 = f(x)$

c. u. $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in[0,1]} |f_n(x) - f(x)| \right)$

Notăm: $g(x) = |f_n(x) - \underbrace{f(x)}_{0}| = nx(1-x)^n ; g : (0,1] \to \mathbb{R}$

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$g'(x) = n(1-x)^n + n^2 x(1-x)^{n-1}\cdot(-1) =$

$= n(1-x)^{n-1}\left(1-x-nx\right) = 0 \Leftrightarrow x = 1 \text{ sau } \frac{1}{n+1}$

x	0		$\frac{1}{n+1}$		1
$g'(x)$	1	$+\ +\ +\ +\ +\ +\ +$	0	$-\ -\ -\ -\ -\ -\ -\ -$	0
$g(x)$	1	↗		↘	

$\Rightarrow \sup_{x\in(0,1]} g(x) = g\left(\frac{1}{n+1}\right) = \frac{n}{n+1}\left(1-\frac{1}{n+1}\right)^n$

$\Rightarrow \lim_{n\to\infty}\left(\sup_{x\in(0,1]} g(x)\right) = \lim_{n\to\infty} \frac{n}{n+1}\left(1-\frac{1}{n+1}\right)^n =$

$= \lim_{n\to\infty} \underbrace{\frac{n}{n+1}}_{\to 1}\Bigg[\underbrace{\left(1+\frac{-1}{n+1}\right)^{-n-1}}_{\to e}\Bigg]^{\frac{-n}{n+1}} = e^{-1} = \frac{1}{e} \neq 0$

$\Rightarrow$ nu e uniform convergentă.

3. $f_n : [0,1] \to \mathbb{R} ; f_n(x) = x^n - x^{2n} ; n \geq 0$

c.s. $\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \underbrace{x^n}_{\to 0}\left(\underbrace{1-x^n}_{\to 1}\right) - 0 = f(x)$

c.u. $\lim_{n\to\infty}\left(\sup_{x\in[0,1]} |f_n(x) - f(x)|\right)$

Notăm: $g(x) = |f_n(x) - \underbrace{f(x)}_{0}| = x^n - x^{2n} = x^n(1-x^n)$

$g : [0,1] \to \mathbb{R}$

$g'(x) = nx^{n-1}(1-x^n) + x^n(-nx^{n-1}) =$

$= nx^{n-1}(1-2x^n) = 0 \Leftrightarrow x = 0 \text{ sau } x = \sqrt[n]{\frac{1}{2}}.$

x	0		$\sqrt[n]{\frac{1}{2}}$		1
$g'(x)$	0	$+\ +\ +\ +\ +\ +\ +$	0	$-\ -\ -\ -\ -\ -\ -\ -$	
$g(x)$		↗		↘	

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$\Rightarrow \sup_{x\in[0,1]} g(x) = g\left(\sqrt[n]{\frac{1}{2}}\right) = \left(\frac{1}{\sqrt[n]{2}}\right)^n \left[1-\left(\frac{1}{\sqrt[n]{2}}\right)^n\right] = \frac{1}{2}\left(1-\frac{1}{2}\right) = \frac{1}{4} \neq 0$

$\Rightarrow$ nu este uniform convergentă

4. $f_n : \mathbb{R} \to \mathbb{R}$; $f_n(x) = \sqrt{x^2+\frac{1}{n^2}}$; $n>0$

C.S.: $\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sqrt{x^2+\underbrace{\frac{1}{n^2}}_{\to 0}} = \sqrt{x^2} = f(x)$

C.U.: $\lim_{n\to\infty} \left(\sup_{x\in\mathbb{R}} |f_n(x) - f(x)|\right)$

Notăm: $g(x) = |f_n(x) - f(x)|$; $g: \mathbb{R}\to\mathbb{R}$;

$g(x) = \sqrt{x^2+\frac{1}{n}} - \sqrt{x^2}$

$g'(x) = \frac{1}{2\sqrt{x^2+\frac{1}{n}}}\cdot 2x - \frac{1}{2\sqrt{x^2}}\cdot 2x = x\underbrace{\left(\frac{1}{\sqrt{x^2+\frac{1}{n}}} - \frac{1}{\sqrt{x^2}}\right)}_{0} = 0 \Leftrightarrow$

$\Leftrightarrow x = 0$

x	$-\infty$		0		∞
$g'(x)$	$+++++++$		0	$-------$	
$g(x)$	$\nearrow$		max	$\searrow$	

$\Rightarrow \sup_{x\in\mathbb{R}} g(x) = g(0) = \sqrt{\frac{1}{n}}$

$\Rightarrow \lim_{n\to\infty}\left(\sup_{x\in\mathbb{R}} g(x)\right) = \lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0 \Rightarrow$

$\Rightarrow$ converge uniform la $f(x) = \sqrt{x^2} = |x|$.

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5. $f_n : (-\infty, 0) \to \mathbb{R}$; $f_n(x) = \frac{e^{nx}-1}{e^{nx}+1}$; $n \geq 0$

C.S.: $\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} \frac{e^{nx}-1}{e^{nx}+1} = -1 = f(x)$

C.U.: $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in(-\infty,0)} |f_n(x) - f| \right)$

Notăm: $g(x) = \left| f_n(x) - f(x) \right| = \left| \frac{e^{nx}-1}{e^{nx}+1} + 1 \right| =$

$= \left| \frac{e^{nx} - 1 + e^{nx} + 1}{e^{nx}+1} \right| = \frac{2e^{nx}}{e^{nx}+1}$; $g : (-\infty, 0) \to \mathbb{R}$

$g'(x) = \frac{2e^{nx} \cdot n (e^{nx}+1) - 2e^{nx}(e^{nx} \cdot n)}{(e^{nx}+1)^2} = \frac{2e^{nx} \cdot n}{(e^{nx}+1)^2} > 0$

x	$-\infty$									0
$g'(x)$	+	+	+	+	+	+	+	+	+	+
$g(x)$										→

$\Rightarrow \sup\limits_{x\in(-\infty,0)} g(x) = \lim\limits_{\substack{x\to 0 \\ x<0}} g(x) = \lim\limits_{\substack{x\to 0 \\ x<0}} \frac{2e^{nx}}{e^{nx}+1} = 1$

$\Rightarrow \lim\limits_{n\to\infty} \left(\sup\limits_{x\in(-\infty,0)} g(x) \right) = \lim\limits_{n\to\infty} 1 = 1 \Rightarrow$

$\Rightarrow$ nu este uniform convergentă

6. $f_n : [-1,1] \to \mathbb{R}$: $f_n(x) = \frac{x}{nx^2+1}$; $n \geq 0$

C.S.: $\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} \frac{x}{nx^2+1} = 0 = f(x)$

C.U.: $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in[-1,1]} |f_n(x) - f(x)| \right)$

Notăm: $g(x) = |f_n(x) - \underbrace{f(x)}_{0}| = \frac{|x|}{nx^2+1}$; $g : [-1,1] \to \mathbb{R}$

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$$g'(x) = \frac{nx^2+1 - x(2nx)}{(nx^2+1)^2} = \frac{1-nx^2}{(nx^2+1)^2} \overset{0}{\iff} x = \frac{1}{\sqrt{n}}$$

x	-1		$\frac{1}{\sqrt{n}}$		1
$g'(x)$	$+++++++++$		0	$---------$	
$g(x)$		$\nearrow$		$\searrow$	

$$\Rightarrow \sup_{x\in[-1,1]} g(x) = g\left(\frac{1}{\sqrt{n}}\right) = \frac{\frac{1}{\sqrt{n}}}{n\cdot\frac{1}{n}+1} = \frac{1}{2\sqrt{n}}$$

$$\Rightarrow \lim_{n\to\infty}\left(\sup_{x\in[-1,1]} g(x)\right) = \lim_{n\to\infty} \frac{1}{2\sqrt{n}} = 0 \Rightarrow$$

$\Rightarrow$ converge uniform la $f(x) = 0$

7. $f_n : \mathbb{R} \to \mathbb{R} : f_n(x) = \operatorname{arctg} \frac{x}{1+n(n+1)x^2}$; $n > 0$

c.s.: $\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \operatorname{arctg} \frac{x}{1+n(n+1)x^2} = 0 = f(x)$

c.u.: $\lim_{n\to\infty} \left(\sup_{x\in\mathbb{R}} |f_n(x) - f(x)|\right)$

Notăm: $g(x) = |f_n(x) - \underbrace{f(x)}_{0}| = \operatorname{arctg} \frac{x}{1+n(n+1)x^2}$;

$g : \mathbb{R} \to \mathbb{R}$.

$$g'(x) = \frac{1}{1+\frac{x^2}{[1+n(n+1)x^2]^2}} \cdot \frac{1+n(n+1)x^2 - x\cdot 2n(n+1)x}{[1+n(n+1)x^2]^2} =$$

$$= \frac{1-n(n+1)x^2}{x^2+[1+n(n+1)x^2]^2} = 0 \iff x = \frac{1}{\sqrt{n(n+1)}}$$

x	$-\infty$		$\frac{1}{\sqrt{n(n+1)}}$		∞
$g'(x)$	$+++++++++$		0	$---------$	
$g(x)$		$\nearrow$		$\searrow$	

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$$\Rightarrow \sup_{x\in\mathbb{R}} g(x) - g\left(\frac{1}{\sqrt{n(n+1)}}\right) = \operatorname{arctg}\frac{\frac{1}{\sqrt{n(n+1)}}}{1+\cancel{n(n+1)}\cdot\frac{1}{\cancel{n(n+1)}}} =$$

$$= \operatorname{arctg}\frac{1}{2\sqrt{n(n+1)}}$$

$$\Rightarrow \lim_{n\to\infty}\left(\sup_{x\in\mathbb{R}} g(x)\right) = \lim_{n\to\infty} \operatorname{arctg}\frac{1}{2\sqrt{n(n+1)}} = 0 \Rightarrow$$

$\rightarrow$ converge uniform la $f(x) = 0$.

8. $f_n : (0,\infty) \to \mathbb{R}$; $f_n(x) = \operatorname{arctg}\frac{x}{n^2}$; $n \geq 1$

c.s.: $\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \operatorname{arctg}\frac{x}{n^2} = 0 = f(x)$

c.u.: $\lim_{n\to\infty}\left(\sup_{x\in(0,\infty)} |f_n(x) - f(x)|\right)$

Notăm: $g(x) = |f_n(x) - f(x)| = \operatorname{arctg}\frac{x}{n^2}$; $g:(0,\infty)\to\mathbb{R}$

$$g'(x) = \frac{1}{1+\frac{x^2}{n^4}}\cdot\frac{n^2-0}{n^4} = \frac{n^2}{n^4+x^2} > 0;\ \forall x\in(0,\infty)$$

x	0	∞
$g'(x)$	+ + + + + + + + + + + + + + + +	
$g(x)$	$\nearrow$	

$$\Rightarrow \sup_{x\in(0,\infty)} g(x) = \lim_{x\to\infty} g(x) = \lim_{x\to\infty} \operatorname{arctg}\frac{x}{n^2} = \frac{\pi}{2}$$

$$\Rightarrow \lim_{n\to\infty}\left(\sup_{x\in(0,\infty)} g(x)\right) = \lim_{n\to\infty}\frac{\pi}{2} = \frac{\pi}{2} \neq 0 \Rightarrow$$

$\Rightarrow$ nu converge uniform.

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2. $f_n : \mathbb{R} \to \mathbb{R}$; $f_n(x) = 2n^2 \cdot x \cdot e^{-n^2x^2}$; $n \geq 1$

c.s.: $\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} 2n^2 \cdot x \cdot e^{-n^2x^2} = \lim\limits_{n\to\infty} \frac{2n^2x}{e^{n^2x^2}}$

Indexăm funcția $a : \mathbb{R} \to \mathbb{R}$; $a(t) = \frac{2t^2x}{e^{t^2x^2}}$, aferentă lui f_n.

$$\lim_{t\to\infty} a(t) = \lim_{t\to\infty} \frac{2t^2x}{e^{t^2x^2}} \overset{\frac{\infty}{\infty}}{\underset{L'Hos}{=}} \lim_{t\to\infty} \frac{4x\cdot t}{e^{t^2x^2}\cdot 2tx^2} =$$

$$= \lim_{t\to\infty} \frac{2}{xe^{t^2x^2}} = 0 \Rightarrow$$

$$\Rightarrow \lim_{n\to\infty} \frac{2n^2x^2}{e^{n^2x^2}} = 0 = f(x)$$

C.U.: $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in\mathbb{R}} |f_n(x) - \underbrace{f(x)}_{0}| \right)$

Notăm: $g(x) = |f_n(x) - \underbrace{f(x)}_{0}| = \begin{cases} x\cdot 2n^2\cdot e^{-n^2x^2} ; x \geq 0 \\ -x\cdot 2n^2\cdot e^{-n^2x^2} ; x < 0 \end{cases}$

$$\Rightarrow g'(x) = \begin{cases} \frac{2n^2}{e^{n^2x^2}}(1-2n^2x^2) ; x \geq 0 \\ -\frac{2n^2}{e^{n^2x^2}}(1-2n^2x^2) ; x < 0 \end{cases} = 0 \Leftrightarrow$$

$\Leftrightarrow 2n^2x^2 = 1 \Leftrightarrow x = \frac{\pm 1}{n\sqrt{2}}$

x	$-\infty$		$\frac{-1}{n\sqrt{2}}$		0		$\frac{1}{n\sqrt{2}}$		∞
$g'(x)$	+	+	0	−		+	0	−	−
$g(x)$	↗		max	↘	0	↗	max	↘	

$$\Rightarrow \sup_{x\in\mathbb{R}} g(x) = g\left(\frac{\pm 1}{n\sqrt{2}}\right) = \frac{n\sqrt{2}}{\sqrt{e}} \Rightarrow$$

$$\Rightarrow \lim_{n\to\infty} \left(\sup_{x\in\mathbb{R}} g(x) \right) = \lim_{n\to\infty} \frac{n\sqrt{2}}{\sqrt{e}} = +\infty \neq 0 \Rightarrow$$

$\Rightarrow$ nu converge uniform.

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10. $f_n : \mathbb{R} \to \mathbb{R}$; $f_n(x) = x + \frac{1}{n}$; $n > 0$. Să se studieze convergența simplă și uniformă pentru $f_n(x)$ și $f_n^2(x)$.

- Pentru $f_n(x)$:

C.S.: $\lim\limits_{n\to\infty} f_n(x) = \lim\limits_{n\to\infty} \left(x + \frac{1}{n}\right) = x = f(x)$

$\Rightarrow f_n \xrightarrow{s} f$

C.U.: $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in\mathbb{R}} |f_n(x) - f(x)|\right)$

Notăm: $g(x) = |f_n(x) - f(x)| = \frac{1}{n}$; $g : \mathbb{R} \to \mathbb{R}$

$\Rightarrow \sup\limits_{x\in\mathbb{R}} g(x) = \frac{1}{n}$

$\Rightarrow \lim\limits_{n\to\infty} \left(\sup\limits_{x\in\mathbb{R}} g(x)\right) = \lim\limits_{n\to\infty} \frac{1}{n} = 0$

$\Rightarrow f_n \xrightarrow{u} f$

- Pentru $f_n^2(x)$:

C.S.: $\lim\limits_{n\to\infty} f_n^2(x) = \lim\limits_{n\to\infty} \left(x + \frac{1}{n}\right)^2 = x^2 = f_1(x)$

$\Rightarrow f_n \xrightarrow{s} f$

C.U.: $\lim\limits_{n\to\infty} \left(\sup\limits_{x\in\mathbb{R}} |f_n^2(x) - f_1(x)|\right)$

Notăm: $h(x) = |f_n^2(x) - f_1(x)| = \left|\left(x + \frac{1}{n}\right)^2 - x^2\right| =$

$= \left|x^2 + 2\cdot\frac{x}{n} + \frac{1}{n^2} - x^2\right| =$

$= 2\frac{x}{n} + \frac{1}{n^2}$; $h : \mathbb{R} \to \mathbb{R}$.

$h'(x) = \frac{2}{n} > 0$; $\forall x \in \mathbb{R}$

x	$-\infty$																				∞
$h'(x)$	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
$h(x)$																					↗

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$\Rightarrow \sup_{x\in\mathbb{R}} h(x) = \lim_{x\to\infty} h(x) = \lim_{x\to\infty} \left(\frac{2}{n}x + \frac{1}{n^2}\right) = +\infty$

$\Rightarrow f_n \not\rightrightarrows f$

11. Fie $\sum_{n\geq 1} n^{-x}$; $x \in \mathbb{R}$. Studiați convergența simplă, convergența uniformă și dacă se poate deriva termen cu termen.

C.S.: $\sum_{n\geq 1} n^{-x} = \sum_{n\geq 1} \frac{1}{n^x}$, convergentă pt $x \in (1,\infty)$ (serie armonică generalizată)

C.U.: Folosim criteriul lui Weierstrass pe $x \in (1,\infty)$

Fie $u_n = \frac{1}{n^\alpha}$; $\alpha \in \mathbb{R} \Rightarrow |f_n(x)| \leq u_n \Leftrightarrow$

$\Leftrightarrow \frac{1}{n^x} \leq \frac{1}{n^\alpha}$; $x \in (1,\infty)$; $\forall \alpha > 1$ (1)

$\sum_{n\geq 1} u_n$ conv pt $\alpha > 1$ (2)

(1)(2) $\Rightarrow \sum_{n\geq 1} n^{-x}$ uniform convergentă pe $[\alpha,\infty)$; $\alpha > 1$.

$f_n(x)$ derivabilă (operații cu funcții elementare)

$\Rightarrow f_n(x)$ este de clasă C^1 (1)

Am arătat că $\sum_{n\geq 1} f_n \to f$ (2)

$\sum_{n\geq 1} f_n' = -\sum_{n\geq 1} n^{-x} \ln n$

Trebuie să arătăm că $\sum_{n\geq 1} n^{-x}\ln n \to g$

Folosim criteriul lui Weierstrass:

$|n^{-x}\ln n| \leq n^{-x}\cdot n = \frac{1}{n^{x-1}}$

$\sum_{n\geq 1} \frac{1}{n^{x-1}}$ convergentă pt $x > 2 \Rightarrow \sum_{n\geq 1} f_n' \to g$ (3)

(1)(2)(3) se poate deriva termen cu termen.

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12. Fie $\sum_{n\geq 1}(-1)^n \frac{e^{-nx}\sqrt{n}}{n}$; $x\in\mathbb{R}^+$. Studiați convergența simplă și absolut convergența.

C.S.: $\sum_{n\geq 1}(-1)^n \frac{e^{-nx}\sqrt{n}}{n}$; $x_n=(-1)^n\frac{e^{-nx}\sqrt{n}}{n}$; $a_n=\frac{e^{-nx}\sqrt{n}}{n}$

Folosim criteriul Leibnitz:

$$\frac{a_{n+1}}{a_n}=\frac{e^{-(n+1)x}\sqrt{n+1}}{n+1}\cdot\frac{n}{e^{-nx}\cdot\sqrt{n}}=\frac{e^{-(n+1)x}}{e^{-nx}}\cdot\sqrt{\frac{n}{n+1}}=$$

$$=\frac{e^{nx}}{e^{nx}\cdot e^{x}}\cdot\sqrt{\frac{n}{n+1}}=\frac{1}{e^x}\cdot\sqrt{\frac{n}{n+1}}<1\Rightarrow$$

$\Rightarrow a_{n+1}<a_n\Rightarrow a_n\searrow$ (1)

$$\lim_{n\to\infty}a_n=\lim_{n\to\infty}\frac{e^{-nx}\sqrt{n}}{n}=\lim_{n\to\infty}\frac{1}{e^{nx}\cdot\sqrt{n}}=0 \quad (2)$$

$\overset{(1)(2)}{\Rightarrow}\sum_{n\geq 1}x_n$ convergentă $\Rightarrow\sum_{n\geq 1}f_n\xrightarrow{S}f$

C.A.: $\sum_{n\geq 1}\left|(-1)^n\frac{e^{-nx}\sqrt{n}}{n}\right|=\sum_{n\geq 1}\frac{e^{-nx}\sqrt{n}}{n}$

Notăm: $y_n=\frac{e^{-nx}\sqrt{n}}{n}=\frac{1}{e^{nx}\sqrt{n}}$

Pentru a stabili natura seriei $\sum_{n\geq 1}y_n$ folosim criteriul raportului:

$$\lim_{n\to\infty}\frac{y_{n+1}}{y_n}=\lim_{n\to\infty}\frac{e^{nx}\sqrt{n}}{e^{(n+1)x}\sqrt{n+1}}=\lim_{n\to\infty}\frac{1}{e^x}\underbrace{\sqrt{\frac{n}{n+1}}}_{\to 1}=\frac{1}{e^x}<1 \quad \forall x\in\mathbb{R}^+$$

$\Rightarrow\sum_{n\geq 1}y_n$ convergentă

$\Rightarrow\sum_{n\geq 1}f_n$ absolut convergentă.

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